

The ODMAP Protocol

A Standard Protocol for Reporting Species Distribution Models for
Reproducibility, Transparency, and Evaluation

[https://nsojournals.onlinelibrary.wiley.com/doi/full/10.1111/
ecog.04960](https://nsojournals.onlinelibrary.wiley.com/doi/full/10.1111/ecog.04960)

SDM 的繁荣与背后的可复现危机

SDM 应用的爆炸式增长

主流工具：物种分布模型已成为生态学、演化生物学与保护生物学中最广泛使用的工具，年发文量超过 1000 篇。

政策依赖：广泛应用于 IPBES 全球生物多样性评估、IUCN 濒危物种红色名录及气候变化风险预测。

门槛降低：易用软件与海量数字地理数据的普及加速了应用，但也导致“黑箱化”操作泛滥。

报告规范的严重缺失

方法不可重现：大量文献缺乏对数据清洗、变量筛选、参数调整与不确定性的完整记录，无法独立复现。

信息分散割裂：建模关键细节常散落在正文、附录各处甚至完全遗漏，极大阻碍同行评议与元分析。

决策风险：不严谨的建模与不透明的报告可能导致错误的生态推断，进而误导关键的保护决策。

实证：近半数 SDM 研究存在缺陷

46%

SDM 研究存在关键缺陷

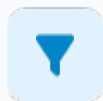
Araújo et al. (2019) 400 篇文献抽样评估

在对 400 篇已发表的物种分布模型研究评估中，**46%** 的论文在数据、算法、验证或预测维度至少存在一项严重缺陷。

高频缺失项包括：未检验环境变量的不确定性、缺乏模型结构与参数敏感性测试、忽视核心生态假设的成立边界。

根本根源：缺乏被学界共同遵守的标准报告框架（Standard Reporting Protocol），导致研究者“无章可循”。

传统 SDM 报告的盲区



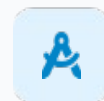
数据偏差处理

采样偏差（Sampling Bias）与不完美探测（Imperfect Detection）未说明校正方法，伪缺失 / 背景点采样策略含糊不清。



算法与参数设置

仅简单提及“使用默认参数”，未注明具体软件版本、正则化乘数、特征类型或 MCMC 链收敛诊断准则。



生态假设检验

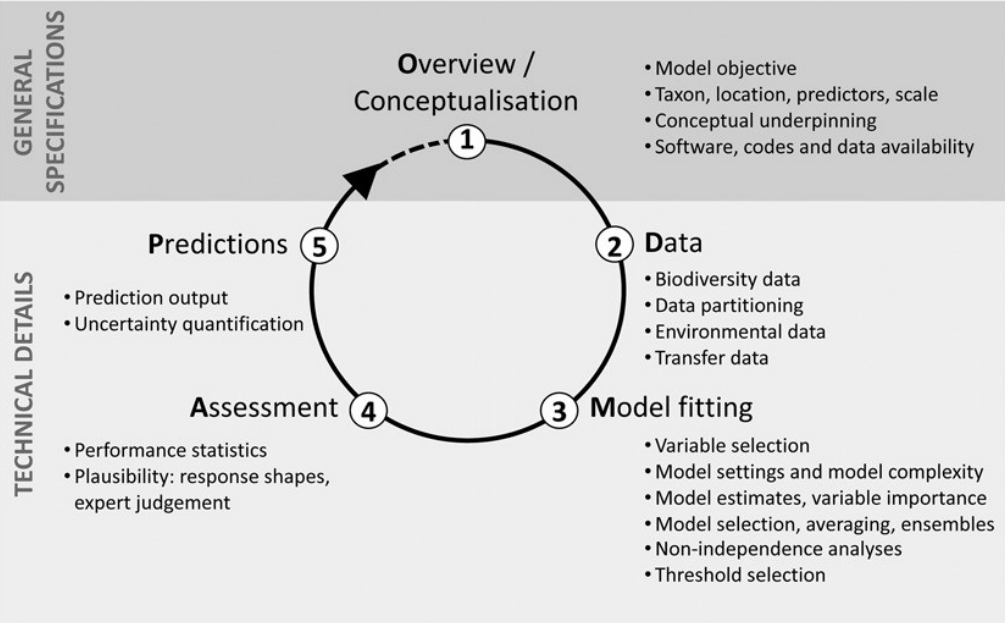
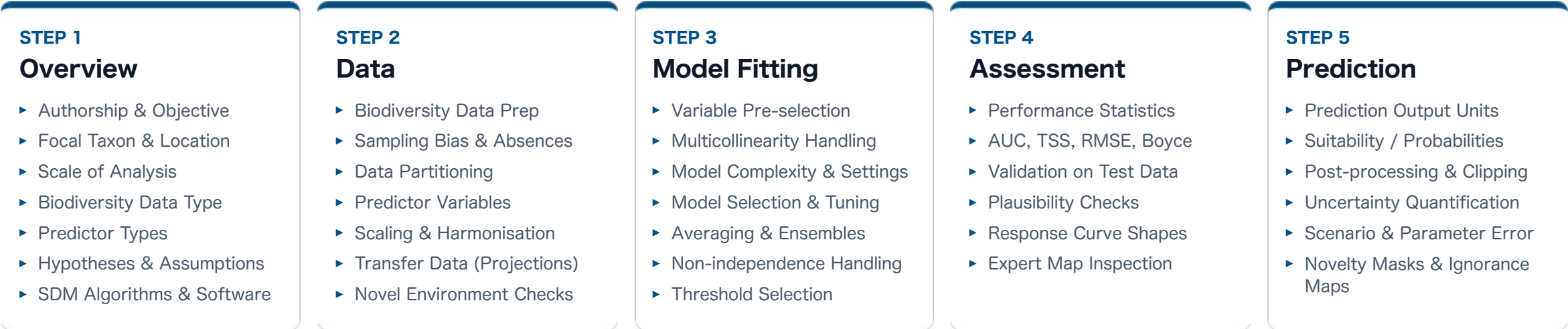
忽略物种 - 环境平衡假设、生态位保守性假设以及空间自相关（Spatial Autocorrelation）对残差的干扰。



外推与不确定性

跨时空投影时未量化新环境（Novel Environments），缺乏预测不确定性地图（Maps of Ignorance）。

ODMAP 端到端 workflow



ODMAP section	ODMAP subsection	ODMAP elements
Overview	Authorship	Authors, contact email, title, doi
	Model objective/model purpose	SDM objective/purpose (inference, mapping, transfer), main target output
	Taxon	Focal taxon
	Location	Location of study area
	Scale of analysis	Spatial extent (lon/lat), spatial resolution, temporal extent/time period, temporal resolution, type of extent boundary (e.g. rectangular, natural, political)
Data	Biodiversity data overview	Observation type, response/data type
	Type of predictors	Climatic, topographic, edaphic, habitat, etc.
	Conceptual model/hypotheses	Hypotheses about biodiversity-environment relationships
	Assumptions	State critical model assumptions (cf. Table 2)
	SDM algorithms	Model algorithms, justification of model complexity, is model averaging/ensemble modelling used?
Model	Model workflow	Brief description of modelling steps
	Software, codes and data	Specify software, availability of codes, availability of data
	Biodiversity data	Taxon names, taxonomic reference system, ecological level, biodiversity data sources, sampling design, sample size per taxon, country/region mask, details on scaling, data cleaning/filtering, absence data collection, pseudo-absence and background data, potential errors and biases in data
	Data partitioning	Selection of training data (for model fitting), validation data and test (truly independent) data
	Predictor variables	State predictor variables used, data sources, spatial resolution and extent of raw data, map projection, temporal resolution and extent of raw data, data processing and scaling, measurement errors and bias, dimension reduction
Assessment	Transfer data for projection	Data sources, spatial resolution and extent, temporal resolution and extent, models and scenarios used, data processing and scaling, quantification of novel environments
	Variable pre-selection	Details on pre-selection of variables
	Multicollinearity	Methods for identifying and dealing with multicollinearity
	Model settings/model complexity	Models settings for all selected algorithms and for extrapolation beyond sample range
	Model estimates	Model coefficients, variable importance
Prediction	Model selection/model averaging/ensembles	Model selection strategy, method for model averaging, ensemble method
	Non-independence analyses	Spatial autocorrelation in residuals, temporal autocorrelation in residuals, nested data
	Threshold selection	Details on threshold selection
	Performance statistics	Performance statistics estimated on training data, on validation data and on test (truly independent) data
	Plausibility check	Response plots: expert judgements (e.g. map display)
Prediction	Prediction output	Prediction unit: post-processing steps
	Uncertainty quantification	Uncertainty through algorithms, input data, parameters, scenarios; visualisation/treatment of novel environments

■

 Obligatory;

■

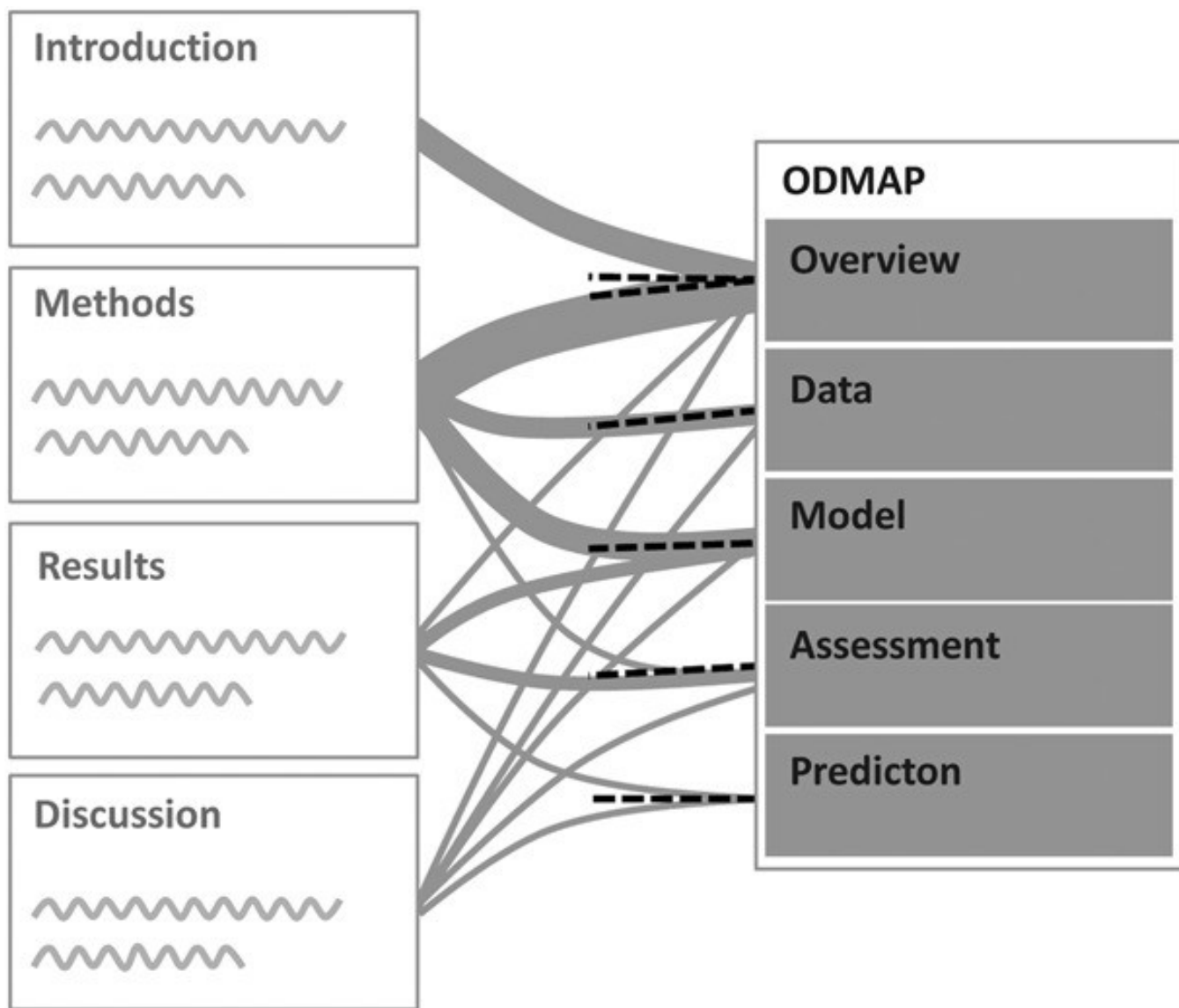
 Objective: mapping/interpolation;

■

 Objective: forecast/transfer;

□

 Optional/context dependent.



ODMAP CHECKLIST

AUTHORS

- ☒ Study design
- ☒ Science writing

REVIEWERS / EDITORS

- ☒ Peer review
- ☒ Author guidelines

EVALUATORS

- ☒ Meta analyses